

Competency

- To solve the quadratic equations

Learning Outcomes

- To solve the quadratic equations by factorization, completing square methods and using quadratic formula

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define quadratic equation - To write the roots of the equation $ax^2+bx+c=0$, $a \neq 0$
2.	Understanding (U)	- To solve the simple quadratic equations by factorization method
3.	Application (A)	- To solve the quadratic equation by factorization method - To solve the quadratic equation by completing square method - To solve the quadratic equation by using formula
4.	High Ability (HA)	- To identify the contextual problems as related to quadratic equations and solve them

Required Teaching Materials/ Resources

Chart paper with various quadratic equations, ICT tools (if possible), audio-video materials etc.

Pre-knowledge: Quadratic equation, factors of quadratic expressions etc

Teaching Activities

1. Write some equation on the board or show in chart paper and ask about the degree and variable
2. Explain about quadratic equations with examples.
3. Recall the factorization and discuss about the solution of quadratic equation by factorization method

For example: $x^2 - 5x + 6 = 0$ or, $x^2 - 3x - 2x + 6 = 0$ or, $(x - 2)(x - 3) = 0$

Either, $x - 2 = 0$ $\therefore x = 2$ or, $x - 3 = 0$ $\therefore x = 3$

Hence, $x = 2$ or 3

4. Give the similar questions for class work and discuss on the solutions
5. To develop critical thinking in the students, give the values of variable and tell to make the quadratic equations satisfying the values and discuss on the solutions again.

6. Recall the formulae $a^2 \pm 2ab + b^2 = (a \pm b)^2$ then discuss on the solution of the quadratic equation by completing square method
7. Discuss about the solution of the equation $ax^2 + bx + c = 0$, $a \neq 0$ by completing the square method

$$ax^2 + bx + c = 0$$

$$\text{or, } ax^2 + bx = -c \quad (\text{Transposing } c \text{ to R.H.S.})$$

$$\text{or, } \frac{ax^2}{a} + \frac{b}{a}x = -\frac{c}{a} \quad (\text{To make the coefficient of } x^2 \text{ unity, dividing both sides by } a)$$

$$\text{or, } x^2 + \frac{b}{a}x = -\frac{c}{a}$$

$$\text{or, } x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = \left(\frac{b}{2a}\right)^2 - \frac{c}{a} \quad (\text{Adding the square of half of the coefficient of } x \text{ to both sides})$$

$$\text{or, } x^2 + 2 \cdot \frac{b}{2a}x + \left(\frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$$

$$\text{or, } \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

$$\text{or, } x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$\text{or, } x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\text{Thus, the required roots of } x \text{ are } \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

8. Solve some quadratic equation by using the above formula and ask some similar equations to be solved by formula.

Solution of selected problems from Vedanta Excel in Mathematics

1. Solve the following equations by factorization method.

$$a) \frac{x+5}{x-5} + \frac{x-5}{x+5} = 2\frac{1}{2}$$

$$b) \frac{x-2}{x+2} + \frac{x+2}{x-2} = \frac{2x+6}{x-3}$$

$$c) \frac{x-2}{x-3} + \frac{3x-11}{x-4} = \frac{4x+13}{x+1}$$

Solution:

$$a) \frac{x+5}{x-5} + \frac{x-5}{x+5} = 2\frac{1}{2}$$

$$\text{or, } \frac{(x+5)^2 + (x-5)^2}{(x-5)(x+5)} = \frac{5}{2}$$

$$\text{or, } \frac{x^2 + 10x + 25 + x^2 - 10x + 25}{x^2 - 25} = \frac{5}{2}$$

$$\text{or, } \frac{2x^2 + 50}{x^2 - 25} = \frac{5}{2}$$

$$\text{or, } 5x^2 - 125 = 4x^2 + 100$$

$$\begin{aligned}
&\text{or, } x^2 = 225 \\
&\text{or, } x^2 - 225 = 0 \\
&\text{or, } x^2 - 15^2 = 0 \\
&\text{or, } (x + 15)(x - 15) = 0 \\
&\text{Either } x + 15 = 0 \quad \therefore x = -15 \\
&\text{or, } x - 15 = 0 \quad \therefore x = 15 \\
&\text{Hence, } x = \pm 15
\end{aligned}$$

$$\begin{aligned}
&\text{b) } \frac{x-2}{x+2} + \frac{x+2}{x-2} = \frac{2x+6}{x-3} \\
&\text{or, } \frac{(x-2)^2(x+2)^2}{(x+2)(x-2)} = \frac{2x+6}{x-3} \\
&\text{or, } \frac{x^2-4x+4+x^2+4x+4}{x^2-4} = \frac{2x+6}{x-3} \\
&\text{or, } \frac{2x^2+8}{x^2-4} = \frac{2x+6}{x-3} \\
&\text{or, } (2x^2+8)(x-3) = (2x+6)(x^2-4) \\
&\text{or, } 2x^3-6x^2+8x-24 = 2x^3-8x+6x^2-24 \\
&\text{or, } -12x^2+16x = 0 \\
&\text{or, } -4x(3x-4) = 0 \\
&\text{Either } -4x = 0 \quad \therefore x = 0 \\
&\text{or, } 3x-4 = 0 \quad \therefore x = \frac{4}{3} \\
&\text{Hence, } x = 0 \text{ or } \frac{4}{3}
\end{aligned}$$

$$\begin{aligned}
&\text{c) } \frac{x-2}{x-3} + \frac{3x-11}{x-4} = \frac{4x+13}{x+1} \\
&\text{or, } \frac{(x-2)(x-4) + (x-3)(3x-11)}{(x-3)(x-4)} = \frac{4x+13}{x+1} \\
&\text{or, } \frac{x^2-4x-2x+8+3x^2-11x-9x+33}{x^2-4x-3x+12} = \frac{4x+13}{x+1} \\
&\text{or, } \frac{4x^2-26x+41}{x^2-7x+12} = \frac{4x+13}{x+1} \\
&\text{or, } 4x^3+4x^2-26x^2-26x+41x+41 = 4x^3-28x^2+48x+13x^2-91x+156 \\
&\text{or, } -7x^2+58x-115 = 0 \\
&\text{or, } 7x^2-58x+115 = 0 \\
&\text{or, } 7x^2-35x-23x+115 = 0 \\
&\text{or, } 7x(x-5)-23(x-5) = 0 \\
&\text{or, } (x-5)(7x-23) = 0 \\
&\text{Either } x-5 = 0 \quad \therefore x = 5 \\
&\text{or, } 7x-23 = 0 \quad \therefore x = \frac{23}{7} \\
&\text{Hence, } x = 5 \text{ or } \frac{23}{7}
\end{aligned}$$

2. Solve these equation by factorization method.

$$a) \sqrt{\frac{x}{x-3}} + \sqrt{\frac{x-3}{x}} = \frac{5}{2}$$

$$b) \frac{2x-3}{x-1} - 4\left(\frac{x-1}{2x-3}\right) = 3$$

Solution:

a) Let $\sqrt{\frac{x}{x-3}} = a$, then $\sqrt{\frac{x-3}{x}} = \frac{1}{a}$

Now, $a + \frac{1}{a} = \frac{5}{2}$

or, $\frac{a^2 + 1}{a} = \frac{5}{2}$

or, $2a^2 + 2 = 5a$

or, $2a^2 - 5a + 2 = 0$

or, $2a^2 - 4a - a + 2 = 0$

or, $2a(a-2) - 1(a-2) = 0$

or, $(a-2)(2a-1) = 0$

Either $a-2 = 0 \quad \therefore a = 2$
or, $2a-1 = 0 \quad \therefore a = \frac{1}{2}$

When $a = 2$, $\sqrt{\frac{x}{x-3}} = 2$

or, $\frac{x}{x-3} = 4$

or, $4x - 12 = x$

or, $x = \frac{12}{3} = 4$

When $a = \frac{1}{2}$, $\sqrt{\frac{x}{x-3}} = \frac{1}{2}$

or, $\frac{x}{x-3} = \frac{1}{4}$

or, $4x = x - 3$

or, $3x = -3 \quad \therefore x = -1$

Hence, $x = -1$ or 4

b) Let $\frac{2x-3}{x-1} = a$, then $\frac{x-1}{2x-3} = \frac{1}{a}$

Now, $a - 4 \times \frac{1}{a} = 3$

or, $\frac{a^2 - 4}{a} = 3$

or, $a^2 - 3a - 4 = 0$

or, $a^2 - 4a + a - 4 = 0$

or, $a(a-4) + 1(a-4) = 0$

or, $(a-4)(a+1) = 0$

Either $a-4 = 0 \quad \therefore a = 4$
or, $a+1 = 0 \quad \therefore a = -1$

When $a = 4$, $\frac{2x-3}{x-1} = 4$

or, $4x - 4 = 2x - 3$

or, $2x = 1 \quad \therefore x = \frac{1}{2}$

When $a = -1$, $\frac{2x-3}{x-1} = -1$

$$\begin{aligned} \text{or, } 2x - 3 &= -x + 1 \\ \text{or, } 3x &= 4 \quad \therefore x = \frac{4}{3} = 1\frac{1}{3} \\ \text{Hence, } x &= \frac{1}{2}, 1\frac{1}{3} \end{aligned}$$

3. Solve these equations.

$$\begin{aligned} \text{a) } \frac{1}{a+b+x} &= \frac{1}{a} + \frac{1}{b} + \frac{1}{x} & \text{b) } \frac{ax^2+bx+c}{px^2+qx+r} &= \frac{ax+b}{px+q} \\ \text{c) } \frac{1}{(a+b)(x-b)} + \frac{1}{(c+a)(x-c)} &= \frac{1}{(x-b)(x-c)} + \frac{1}{(a+b)(c+a)} \end{aligned}$$

Solution:

$$\text{a) } \frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}$$

$$\text{or, } \frac{1}{a+b+x} - \frac{1}{x} = \frac{1}{a} + \frac{1}{b}$$

$$\text{or, } \frac{x-a-b-x}{x(a+b+x)} = \frac{b+a}{ab}$$

$$\text{or, } \frac{-(a+b)}{x(a+b+x)} = \frac{(a+b)}{ab}$$

$$\text{or, } ax+bx+x^2 = -ab$$

$$\text{or, } x^2+ax+bx+ab = 0$$

$$\text{or, } x(x+a)(x+b) = 0$$

$$\text{Either } x+a = 0 \quad \therefore x = -a$$

$$\text{or, } x+b = 0 \quad \therefore x = -b$$

$$\text{Hence, } x = -a \text{ or } -b.$$

$$\text{b) } \frac{ax^2+bx+c}{px^2+qx+r} = \frac{ax+b}{px+q}$$

$$\text{or, } \cancel{apx^3} + \cancel{aqx^2} + \cancel{bp}x^2 + \cancel{bq}x + cp x + cq = \cancel{apx^3} + \cancel{aqx^2} + arx + \cancel{bp}x^2 + \cancel{bq}x + br$$

$$\text{or, } cp x + cq = arx + br$$

$$\text{or, } x(cp-ar) = br-cq$$

$$\therefore x = \frac{br-cq}{cp-ar}$$

$$\text{c) } \frac{1}{(a+b)(x-b)} + \frac{1}{(c+a)(x-c)} = \frac{1}{(x-b)(x-c)} + \frac{1}{(a+b)(c+a)}$$

$$\text{or, } \frac{(c+a)(x-c) + (a+b)(x-c)}{(a+b)(c+a)(x-b)(x-c)} = \frac{(a+b)(c+a) + (x-b)(x-c)}{(a+b)(c+a)(x-b)(x-c)}$$

$$\text{or, } (c+a)(x-c) + (a+b)(x-c) = (a+b)(c+a) + (x-b)(x-c)$$

$$\text{or, } (c+a)(x-c) - (a+b)(c+a) + (a+b)(x-c) - (x-b)(x-c) = 0$$

$$\text{or, } (c+a)(x-c-a-b) + (x-c)(a+b-x+b) = 0$$

$$\text{or, } (x-a-b-c)(a+b+c-x) = 0$$

$$\text{Either } x-a-b-c = 0 \quad \therefore x = a+b+c$$

$$\text{or, } a+b+c-x = 0 \quad \therefore x = a+b+c$$

$$\text{Hence, } x = a+b+c, a+b+c$$

4. Solve each equation by completing the square.

$$\text{a) } 3x^2 + 5x - 2 = 0 \quad \text{b) } 15x^2 - 2ax = a^2$$

Solution:

$$\text{a) } 3x^2 + 5x - 2 = 0$$

$$\text{or, } \frac{3x^2}{3} + \frac{5}{3}x = \frac{2}{3} \quad [\text{Dividing both sides by 3}]$$

$$\text{or, } x^2 + \frac{5}{3}x = \frac{2}{3}$$

$$\text{or, } x^2 + \frac{5}{3}x + \left(\frac{5}{6}\right)^2 = \frac{2}{3} + \left(\frac{5}{6}\right)^2$$

$$\text{or, } x^2 + 2 \cdot x \cdot \frac{5}{6} + \left(\frac{5}{6}\right)^2 = \frac{2}{3} + \frac{25}{36}$$

$$\text{or, } \left(x + \frac{5}{6}\right)^2 = \frac{24 + 25}{36}$$

$$\text{or, } \left(x + \frac{5}{6}\right)^2 = \left(\pm \frac{7}{6}\right)^2$$

Taking (+) ve sign

Taking (-) ve sign

$$x + \frac{5}{6} = \frac{7}{6}$$

$$x + \frac{5}{6} = \frac{-7}{6}$$

$$\therefore x = \frac{2}{6} = \frac{1}{3}$$

$$\therefore x = \frac{-12}{6} = \frac{5}{6}$$

Hence $x = \frac{1}{3}$ or -2 .

$$\text{b) } 15x^2 + 2ax = a^2$$

Dividing each term by 15

$$\frac{15x^2}{15} - \frac{2ax}{15} = \frac{a^2}{15}$$

$$\text{or, } x^2 - \frac{2ax}{15} + \left(\frac{2a}{2 \times 15}\right)^2 = \frac{a^2}{15} + \left(\frac{2a}{2 \times 15}\right)^2$$

$$\text{or, } x^2 - 2 \cdot x \cdot \frac{a}{15} + \left(\frac{a}{15}\right)^2 = \frac{a^2}{15} + \frac{a^2}{225}$$

$$\text{or, } \left(x - \frac{a}{15}\right)^2 = \frac{16a^2}{225}$$

$$\text{or, } \left(x - \frac{a}{15}\right)^2 = \left(\pm \frac{4a}{15}\right)^2$$

Taking (+) ve sign

Taking (-) ve sign

$$x - \frac{a}{15} = \frac{4a}{15}$$

$$x - \frac{a}{15} = \frac{-4a}{15}$$

$$\therefore x = \frac{5a}{15} = \frac{a}{3}$$

$$\therefore x = \frac{-3a}{15} = \frac{-a}{5}$$

Hence $x = \frac{a}{3}$ or $\frac{-a}{5}$.

5. Express $2x^2 - 4x + 5$ in the form of $a(x - h)^2 + k$, where a , h and k are whole numbers. Hence find the root of $x^2 + hx - ak = 0$

Solution:

$$\begin{aligned} \text{Here, } 2x^2 - 4x + 5 &= 2(x^2 - 2x) + 5 \\ &= 2(x^2 - 2 \cdot x \cdot 1 + 1^2 - 1^2) + 5 \\ &= 2\{(x - 1)^2 - 1\} + 5 \\ &= 2(x - 1)^2 - 2 + 5 \\ &= 2(x - 1)^2 + 3 \end{aligned}$$

Which is the form $a(x - h)^2 + k$, where $a = 2$, $h = 1$ and $k = 3$.

Again,

$$x^2 + hx - ak = 0$$

$$\text{or, } x^2 + 1 \cdot x - 2 \times 3 = 0$$

$$\text{or, } x^2 + x - 6 = 0$$

$$\text{or, } x^2 + 3x - 2x - 6 = 0$$

$$\text{or, } x(x + 3) - 2(x + 3) = 0$$

$$\text{or, } (x + 3)(x - 2) = 0$$

Either

$$x + 3 = 0 \quad \therefore x = -3$$

$$\text{or, } x - 2 = 0 \quad \therefore x = 2$$

Hence, the required roots of $x^2 + hx - ak = 0$ are -3 and 2 when $h = 1$, $a = 2$ and $k = 3$.

6. Solve: $px^2 + qx + r = 0$

Solution:

Here, $px^2 + qx + r = 0$

Dividing each term by p we get,

$$\frac{px^2}{p} + \frac{qx}{p} + \frac{r}{p} = \frac{0}{p}$$

$$\text{or, } x^2 + \frac{q}{p}x + \frac{r}{p} = 0$$

$$\text{or, } x^2 + \frac{q}{p}x + \left(\frac{q}{2p}\right)^2 = \frac{-r}{p} + \left(\frac{q}{2p}\right)^2$$

$$\text{or, } \left(x + \frac{q}{2p}\right)^2 = \frac{-r}{p} + \frac{q^2}{4p^2}$$

$$\text{or, } \left(x + \frac{q}{2p}\right)^2 = \frac{q^2 - 4pr}{4p^2}$$

$$\text{or, } x + \frac{q}{2p} = \pm \sqrt{\frac{q^2 - 4pr}{4p^2}}$$

$$\text{or, } x = \frac{-q}{2p} \pm \frac{\sqrt{q^2 - 4pr}}{2p}$$

$$\therefore x = \frac{-q \pm \sqrt{q^2 - 4pr}}{2p}$$

7. Solve the equation $3x^2 + 10 = -11x$ by using formula.

Solution:

Here, $3x^2 + 10 = -11x$

$$\text{or, } 3x^2 + 11x + 10 = 0$$

Comparing it with $ax^2 + bx + c = 0$. we get $a = 3$, $b = 11$ and $c = 10$

We have

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-11 \pm \sqrt{11^2 - 4 \times 3 \times 10}}{2 \times 3} = \frac{-11 \pm \sqrt{1}}{6} = \frac{-11 \pm 1}{6}$$

Taking (+) ve sign,

$$x = \frac{-11 + 1}{6}$$

$$\therefore x = \frac{-5}{3}$$

$$\text{Hence, } x = \frac{-5}{3} \text{ or } -2.$$

Taking (-) ve sign,

$$x = \frac{-11 - 1}{6}$$

$$x = -2$$

$$c) \frac{(x-1)}{(x+1)} + \frac{(x+2)}{(x-2)} = \frac{(x-3)}{(x+3)} + \frac{(x+4)}{(x-4)}$$

Solution:

$$\text{Here, } \frac{(x-1)}{(x+1)} + \frac{(x+2)}{(x-2)} = \frac{(x-3)}{(x+3)} + \frac{(x+4)}{(x-4)}$$

$$\text{or, } \frac{(x-1)(x-2) + (x+2)(x+1)}{(x+1)(x-2)} = \frac{(x-3)(x-4) + (x+4)(x+3)}{(x+3)(x-4)}$$

$$\text{or, } \frac{x^2 - 3x + 2 + x^2 + 3x + 2}{x^2 - x - 2} = \frac{x^2 - 7x + 12 + x^2 + 7x + 12}{x^2 - x - 12}$$

$$\text{or, } \frac{2x^2 + 4}{x^2 - x - 2} = \frac{2x^2 + 24}{x^2 - x - 12}$$

$$\text{or, } \frac{2(x^2 + 2)}{x^2 - x - 2} = \frac{2(x^2 + 12)}{x^2 - x - 12}$$

$$\text{or, } (x^2 + 2)(x^2 - x - 12) = (x^2 + 12)(x^2 - x - 2)$$

$$\text{or, } x^4 - x^3 - 12x^2 + 2x^2 - 2x - 24 = x^4 - x^3 - 2x^2 + 12x^2 - 12x - 24$$

$$\text{or, } -10x^2 - 2x = 10x^2 - 12x$$

$$\text{or, } -20x^2 + 10x = 0$$

$$\text{or, } -10x(2x - 1) = 0$$

$$\text{Either } -10x = 0 \quad \therefore x = 0$$

$$\text{or, } 2x - 1 = 0 \quad \therefore x = \frac{1}{2}$$

$$\text{Hence, } x = 0, \frac{1}{2}$$

$$9. \text{ Solve: } x^2(a-b) - x(a+b) + 2b = 0$$

Solution:

$$\text{Here, } x^2(a-b) - x(a+b) + 2b = 0$$

$$\text{or, } ax^2 - bx^2 - ax - bx + 2b = 0$$

$$\text{or, } ax^2 - ax - bx^2 + bx - 2bx + 2b = 0$$

$$\text{or, } ax(x-1) - bx(x-1) - 2b(x-1) = 0$$

$$\text{or, } (x-1)(ax - bx - 2b) = 0$$

$$\text{Either } x-1 = 0 \quad \therefore x = 1$$

$$\text{or, } ax - bx - 2b = 0$$

$$\text{or, } x(a-b) = 2b \quad \therefore x = \frac{2b}{a-b}$$

$$\text{Hence, } x = 1, \frac{2b}{a-b}$$

Extra Questions

1. **Solve:**

$$a) \frac{x^2 - 1}{5} = 3 \text{ Ans : } \pm 4$$

$$b) \frac{x}{9} = \frac{4}{x} \quad \text{Ans : } \pm 6$$

$$c) 6 - 5x + x^2 = 0 \text{ Ans : } 2, 3$$

2. **Solve:**

$$a) \frac{x+2}{x} - \frac{x}{x+2} = 1\frac{1}{2} \text{ Ans : } 2, -\frac{4}{3} \quad b) \frac{x}{x+1} + \frac{x+1}{x} = 2\frac{1}{6} \text{ Ans : } 2, -3$$

$$c) \frac{x-3}{x-2} + \frac{x+3}{x+2} = \frac{2x-3}{x-1} \text{ Ans : } 0, 4$$